1 Clarify

Confirmed request

Select ≤ 2 sites.
Maximize coverage.

Public: sites, demand
Fixed: budget, radius
Output: selected IDs

Clarifier
Facts + settings

2 Model

Typed binary formulation

x_i: site selected
y_j: demand covered

$$\max \Sigma_j w_j y_j$$

$$y_j \le \Sigma_i a_{ji} x_i$$

$$\Sigma_i x_i \le 2;\ x_i, y_j \in \{0,1\}$$

Optional pattern retrieval

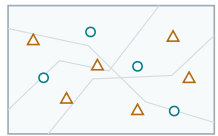
Modeler
Schema + builder

3 Prepare

Public sources → instance

site	lat	lon
s_1	…	…
s_2	…	…

Demand + radius → a_{ji}



Data Scout + Adapter

4 Solve

QUBO → binary candidate

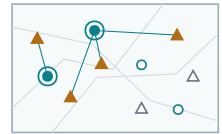
$$E = f + \Sigma_k \lambda_k P_k$$

x = (1, 1, 0, 0)

Compiler + backend
Preflight + bounded calls

5 Deliver

Original-constraint checks



● Selected site
▲ Covered demand

Visualizer + explanation
Decision + provenance

Request + data schema

Requirement Checker
Separate feasibility / ranking evaluator

Data Checker
Independent counts / totals

Delivery status
Usable / limited / unavailable

Validation evidence → bounded data or model repair

Schematic coverage example; checkers provide evidence, not a correctness guarantee.